

Section 7.2 Exercises

Directions: For exercises 1 - 6, find the exact value.

1. $\cos\left(\frac{7\pi}{12}\right)$

2. $\cos\left(\frac{\pi}{12}\right)$

3. $\sin\left(\frac{5\pi}{12}\right)$

4. $\sin\left(\frac{11\pi}{12}\right)$

5. $\tan\left(-\frac{\pi}{12}\right)$

6. $\tan\left(\frac{19\pi}{12}\right)$

Directions: For exercises 7 - 10, rewrite in terms of $\sin x$ and $\cos x$. Simplify all answers.

7. $\sin\left(x + \frac{11\pi}{6}\right)$

8. $\sin\left(x - \frac{3\pi}{4}\right)$

9. $\cos\left(x - \frac{5\pi}{6}\right)$

10. $\cos\left(x + \frac{2\pi}{3}\right)$

Directions: For exercises 11 - 16, simplify the given expression.

11. $\csc\left(\frac{\pi}{2} - t\right)$

12. $\sec\left(\frac{\pi}{2} - \theta\right)$

13. $\cot\left(\frac{\pi}{2} - x\right)$

14. $\tan\left(\frac{\pi}{2} - x\right)$

15. $\sin(2x)\cos(5x) - \sin(5x)\cos(2x)$

16. $\frac{\tan\left(\frac{3}{2}x\right) - \tan\left(\frac{7}{5}x\right)}{1 + \tan\left(\frac{3}{2}x\right)\tan\left(\frac{7}{5}x\right)}$

Directions: For exercises 17 - 18, find the requested information.

17. Given that $\sin a = \frac{2}{3}$ and $\cos b = -\frac{1}{4}$, with a and b both in the interval $\left[\frac{\pi}{2}, \pi\right)$, find $\sin(a + b)$ and $\cos(a - b)$.

18. Given that $\sin a = \frac{4}{5}$, and $\cos b = \frac{1}{3}$, with a and b both in the interval $\left[0, \frac{\pi}{2}\right)$, find $\sin(a - b)$ and $\cos(a + b)$.

Directions: For exercises 19 - 21, find the exact value of each expression.

19. $\sin\left(\cos^{-1}(0) - \cos^{-1}\left(\frac{1}{2}\right)\right)$

20. $\cos\left(\cos^{-1}\left(\frac{\sqrt{2}}{2}\right) + \sin^{-1}\left(\frac{\sqrt{3}}{2}\right)\right)$

21. $\tan\left(\sin^{-1}\left(\frac{1}{2}\right) - \cos^{-1}\left(\frac{1}{2}\right)\right)$

Directions: For exercises 22 - 26, find the exact value.

22. $\sin(75^\circ)$

23. $\sin(195^\circ)$

24. $\cos(165^\circ)$

25. $\cos(345^\circ)$

26. $\tan(-15^\circ)$